(eigenvalues and eigenvectors)

Def:

$$Ax = \lambda x$$

A eigenvector is a special vector that when you multiply a matrix to it, it does not change the line it is on.

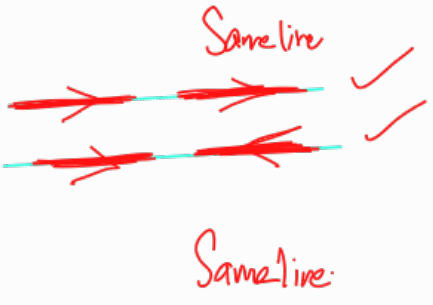
→ The vector only gets longer / shorter / flips

eigenvalue (λ) → tells how much a eigenvector stretches / shrinks.

$\lambda = 3$ vector becomes 3 times longer

$\lambda = 0.5$ vector half its length

$\lambda = -1$ vector flips



$Ax = \lambda x$ (λ: scaled by some value)

- A: matrix
- x: vector
- λx: same vector

Matrix A changes the vector x into a vector that still is along the same line. Just multiplied by A

eigenvector is always non-zero and will always remain on the same line



eg $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$ $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$Ax = \lambda x$

2×2

$$\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

2×1

finding λ =?

$(1,2)$

$(3,0)$ $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$

$(8,-1)$

$$Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 6 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$\boxed{\lambda = 3}$

↳ eigenvalue

(characteristic equation) → used to find eigenvalues (λ) of a matrix A

$$\det(\lambda I - A) = 0$$

↳ In this we are given only matrix A and not eigen vector

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

find λ.

$$(\lambda I - A) \rightarrow \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \ell-3 & 0 \\ -8 & \ell+1 \end{bmatrix} \rightarrow \det \begin{bmatrix} \ell-3 & 0 \\ -8 & \ell+1 \end{bmatrix}$$

$$\rightarrow (\ell-3)(\ell+1) = 0 \qquad \boxed{\ell = 3 \;,\; \ell = -1}$$

eigenvalues

now finding eigenvectors for each eigenvalues

$$\det(\ell I - A) = 0$$

For $\ell = 3$

$$Ax = \ell x$$

$$Ax - \ell x = 0$$

$$x(A - \ell I) = 0$$

$$x(A - 3I) = 0$$

$$\rightarrow \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 8 & -4 \end{bmatrix}$$

$$x \begin{bmatrix} 0 & 0 \\ 8 & -4 \end{bmatrix} = 0 \rightarrow (x) \begin{bmatrix} 0 & 0 \\ 8 & -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$8x_1 - 4x_2 = 0$$

$$x_2 = 2x_1$$

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Now for $\lambda = -1$ $\quad Ax = \lambda x$

$$Ax - \lambda x = 0$$

$$x(A - \lambda I) = 0$$

$$x(A + I) = 0$$

$$\left(\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

$$x \begin{bmatrix} 4 & 0 \\ 8 & 0 \end{bmatrix} = 0$$

$$4x_1 = 0 \qquad 8x_1 = 0$$

$$x_1 = 0 \qquad x_1 = 0$$

$$x = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

as x_2 didn't exists we use on our value. Now eigen vectors must be non-zero

Upper Triangular matrix

$$\begin{bmatrix} a & n & x \\ 0 & b & y \\ 0 & 0 & c \end{bmatrix}$$

Lower triangular matrix

$$\begin{bmatrix} a & 0 & 0 \\ n & b & 0 \\ x & y & c \end{bmatrix}$$

all entries above are 0.

Diagonal matrix

$$\begin{bmatrix} a & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & J \end{bmatrix}$$

If a matrix is diagonal, lower triangular, upper triangular then its eigenvalues are the no of main diagonals

$$A = \begin{bmatrix} 2 & 5 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & -1 \end{bmatrix}$$

$\lambda = 2 \quad \lambda = 3 \quad \lambda = -1$

3x3 eigenvalues

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix} = 0$$

$$\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix} = 0$$

$$\det \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -4 & 17 & \lambda - 8 \end{bmatrix} \Rightarrow \lambda^3 - 8\lambda^2 + 17\lambda - 4$$

$e^3 - 8e^2 + 17e - 4 = 0$ $\quad e = 4,\ e = 3.732,\ e = 0.268$

Set of all sol of $x(eI - A) = 0$ is called eigen space

eigen space contains

→ all eigen vectors

→ zero vector (even though 0 itself is not a eigen vector)

eigen space ↓

or all vectors that satisfy $Ax = \ell x$

ℓ_1, ℓ_2 note:- lamda values should not be repeated

find Bases for the eigenspaces of the matrix.

$$A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix} \qquad \det(\ell I - A) = 0$$

$$\begin{bmatrix} e & 0 \\ 0 & e \end{bmatrix} - \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} e+1 & -3 \\ -2 & e \end{bmatrix}$$

$$\det \begin{pmatrix} e+1 & -3 \\ -2 & e \end{pmatrix} = (e-2)(e+3) = 0$$

$e = 2, \quad e = -3$

find eigen vectors for $e = 2$

$$Ax = \ell x$$

$$Ax - \ell x = 0$$

$$n(A - I\ell) = 0$$

$$n\left(\begin{pmatrix} -1 & 3 \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} \ell & 0 \\ 0 & \ell \end{pmatrix}\right) = 0 \qquad A = \begin{pmatrix} -1 & 3 \\ 2 & 0 \end{pmatrix}$$

$$\left[\begin{pmatrix} -1-\ell & 3 \\ 2 & -\ell \end{pmatrix}\right] \rightarrow \begin{pmatrix} -3 & 3 \\ 2 & -2 \end{pmatrix}$$

$$n\begin{pmatrix} -3 & 3 \\ 2 & -2 \end{pmatrix} = 0$$

$$-3n_1 + 3n_2 = 0 \qquad 3n_2 = 3n_1$$

$$\boxed{n_1 = n_2}$$

$$2n_1 - 2n_2 = 0$$

$$2n_2 - 2n_2 = 0$$

$$0 = 0$$

$$n_1 = t \qquad n_2 = t \qquad t \neq 0$$

$$n = \begin{bmatrix} t \\ t \end{bmatrix} \rightarrow \text{eigen vector}$$

find eigen vector for $\ell = -3$

$$An = \ell n$$

$$An - \ell n = 0$$

$$n(A - I\ell) = 0$$

$$\begin{pmatrix} -1 & 3 \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix} = 0$$

$$n\begin{pmatrix} 2 & 3 \\ 2 & 3 \end{pmatrix} = 0$$

$$2n_1 + 3n_2 = 0$$

$$2n_1 = -3n_2$$

$$\boxed{n_1 = \frac{-3n_2}{2}}$$

$$2n_1 + 3n_2 = 0$$

$$2x\left(-\frac{3x_2}{2}\right) + 3x_2 = 0$$

$$-3x_2 + 3x_2 = 0$$

$$0 = 0$$

$$x = \begin{bmatrix} -\frac{3}{2}x_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2}q \\ q \end{bmatrix} \Rightarrow q\begin{bmatrix} -\frac{3}{2} \\ 1 \end{bmatrix} \qquad x_2 = q$$

e.g 7

$$x = \begin{bmatrix} -s \\ t \\ s \end{bmatrix} \rightarrow \text{eigen vector}$$

$$x = s\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + t\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Basis of this eigen vector

Splitting into variables

If a (lamda) $\ell = 0$ → then Matrix A not Invertible

(Diagonalization) $B = P^{-1}AP$

A is similar to B is there is a invertable matrix P

A and B are Similar means there eigen value is equal

new sys ← $B = P^{-1}AP$

- P^{-1}: convert back from old → new
- A: applies old trans...
- P: converts vector from new coordinate sys to old

A square matrix is diagonizable if you find some matrix P

diagonal ← $D = P^{-1}AP$

Diagonizable → A matrix that can be made a diagonal in some other coordinate sys.

Steps of diagonalizing a matrix: $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

1) find independent eigenvectors

2) Form matrix P using those eigenvectors put them as columns in matrix.

3) compute $P^{-1}AP \rightarrow$ and we get $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

find matrix P that diagonalizes A

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

$$\rightarrow \det\begin{pmatrix} -\lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{pmatrix} \rightarrow (\lambda-1)(\lambda-2)^2 = 0$$

$\lambda = 1 \quad \lambda = 2$

for $\lambda = 1$

$$Ax = \lambda x$$

$$Ax - \lambda x = 0$$

$$x(A - \lambda I) = 0$$

$$\begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$-x_1 - 2x_3 = 0$

$x_1 + x_2 + x_3 = 0$

$x_1 + 2x_3 = 0$

$u_1 = -2u_3$

$-2u_3 + 2u_3 = 0$

$0 = 0$ $\quad$ $\boxed{u_3 = s}$

$$u = \begin{bmatrix} -2s \\ s \\ s \end{bmatrix} \rightarrow s \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$s \neq 0$

eigenvector

Suppose

$\boxed{u_1 = -2s}$

$-2s + u_2 + s = 0$

$-s + u_2 = 0$

$\boxed{u_2 = s}$

for $\ell = 2$

$$\begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$-2u_1 - 2u_3 = 0$

$u_1 + u_3 = 0$

$u_1 + u_3 = 0$

$\boxed{u_1 = -u_3}$ $\quad$ let $u_2 = q$

$-2(-u_3) - 2u_3 = 0$

$2u_3 - 2u_3 = 0$

$0 = 0$ $\quad$ let $u_3 = r$

$u_1 + r = 0$

$u_1 = -r$

$$u = \begin{bmatrix} -r \\ q \\ r \end{bmatrix} \rightarrow r \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + q \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

here

$$P = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Same thing

Order doesn't matter

To check if P diagonalizes A.

$P^{-1}AP = D$ → diagonalized

↓ determinant.

check if $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{pmatrix}$ is diagonalizable or not

find eigen value

$(e-1)(e-2)^2 = 0$

$e = 1 \quad e = 2$

find eigenvectors for $e = 1$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ -3 & 5 & 1 \end{bmatrix}$$

$x_1 + x_2 = 0$

$\boxed{x_1 = -x_2}$

$-3x_1 + 5x_2 + x_3 = 0$

$-3(-x_2) + 5x_2 + x_3 = 0$

$3x_2 + 5x_2 + x_3 = 0$

$8x_2 + x_3 = 0$

$x_3 = -8x_2$

$x_3 = -8t$

let $x_2 = t$

$$\begin{bmatrix} -t \\ t \\ 8 \end{bmatrix} \rightarrow t \begin{bmatrix} -1 \\ 1 \\ 8 \end{bmatrix}$$

Now for $e = 2$ we got an eigen vector $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

A matrix is only diagonalizable if it has as many independent eigen vectors as its size

3 × 3 → 3 Independent eigen vectors

In the above eg we have only 2 eigen vectors

$\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}$ → Independent eigenvectors

$\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}2\\0\\0\end{bmatrix}$ dependent eigenvectors

Scaled

or if we have three different lamdas (each produces diff eigen vectors) then matrix is diagonalizable

$u_2 - \frac{\langle u_2, v_1\rangle}{\|v_1\|^2} v_1$

$\begin{bmatrix}-1 & -1\\1 & 0\\0 & 1\end{bmatrix}$

orthogonal diagonalize

$P^T A P = D$

ℓ^2 is a eigenvalue for A^2

X is still the eigenvector it remains same

So eigen vector obtained from ℓ is same as eigenvector obtained from ℓ^2.

$$A^k = P^{-1} D^k P$$